

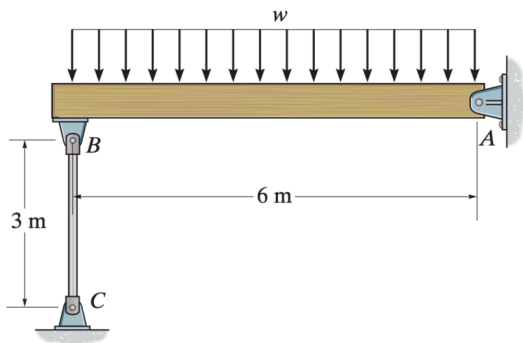
材料系 材料力学B 2022春 期末考试复现

总共六道大题。

仅可体现大致的考试内容与考试风格。

1. 课本670页F13-6

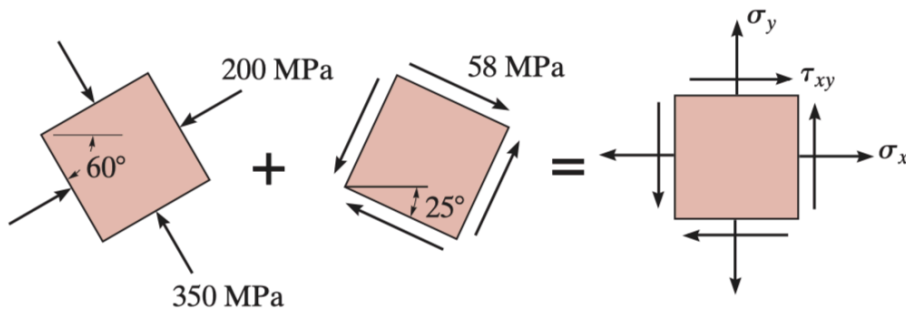
F13-6. The A-36 steel rod BC has a diameter of 50 mm and is used as a strut to support the beam. Determine the maximum intensity w of the uniform distributed load that can be applied to the beam without causing the strut to buckle. Take F.S. = 2 against buckling.



F13-6

2. 课本455页9-18

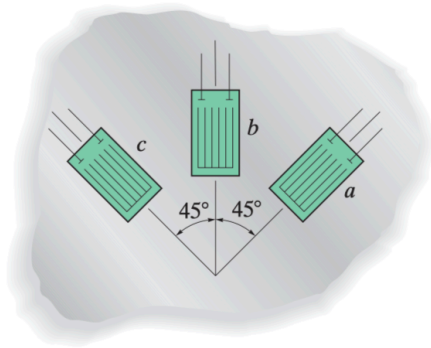
9-18. A point on a thin plate is subjected to the two successive states of stress shown. Determine the resultant state of stress represented on the element oriented as shown on the right.



Prob. 9-18

3. 课本507页10-27, 考试不用画deformed element

10-27. The 45° strain rosette is mounted on a steel shaft. The following readings are obtained from each gauge: $\epsilon_a = 300(10^{-6})$, $\epsilon_b = -250(10^{-6})$, and $\epsilon_c = -450(10^{-6})$. Determine (a) the in-plane principal strains and (b) the maximum in-plane shear strain and average normal strain. In each case show the deformed element due to these strains.

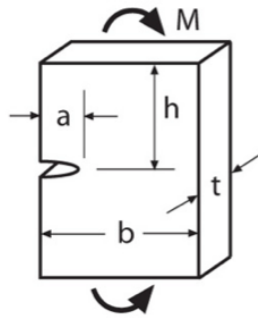


Prob. 10-27

4. 和这一题非常像, 关系式改了。第一问求principal stress里方向2的strain和题给条件关系, 第二问用octahedral shear stress criterion判断yielding。

12.14 Consider plane stress with principal normal stresses σ_1 , $\sigma_2 = \lambda\sigma_1$, and $\sigma_3 = 0$. Assuming a Ramberg–Osgood form of stress–strain curve, derive relationships analogous to Eq. 12.39 for the other two principal strains, ϵ_2 and ϵ_3 , each as a function of σ_1 and λ and materials constants.

5. 长这样的一个bending moment, 给了M的最值 (均值不为0) 和一些参数, 循环做疲劳测试。fracture的内容和fatigue的内容放这个题里面一起考了。F的值不需要用那个很长的公式算, 代固定值就行。



给了safety factor, 用Paris Law计算循环多少圈会fracture。

6. 给了一个S-N curve的参数, safety factor in stress 和 stress amplitude, 求life和life的safety factor, 就是这个图。

